

PROBLEM STATEMENT DETAILS

Problem Statement ID	26110
Problem Statement title	Development of a Low cost Light-Weight Milk Chilling Can for Small-Scale Dairy Farmers
Description	Background Milk is a highly perishable agricultural product that begins to deteriorate rapidly after milking due to bacterial growth and enzymatic activity. In many rural and remote dairy-producing regions, particularly Hilly

	and North Eastern Regions, farmers lack access to bulk milk cooling facilities and reliable electricity. As a result, milk often remains at ambient temperatures for several hours before reaching collection centers, leading to quality degradation, reduced shelf life and economic losses. Available solutions are expensive, relatively heavy and low volume storage cans, making them uneconomical for use or during transportation. Description The proposed
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	problem aims to develop a low-cost, lightweight milk chilling can capable of maintaining milk at safe storage temperatures during collection and transportation. The can should be manufactured using food-grade lightweight materials such as High-Density Polyethylene (HDPE), aluminum alloys or composite materials while ensuring structural strength and hygiene standards. The design should incorporate an insulated double-wall structure
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with materials such as polyurethane foam (PUF) or other cost-effective insulating materials to minimize heat transfer.

The chilling mechanism may utilize reusable ice packs, phase change materials (PCM), or passive cooling technologies that do not require continuous electrical power. The system should be capable of maintaining milk temperatures between 4°C and 8°C for several hours under typical rural

	environmental conditions. The solution should also consider ease of handling, cleaning, transportation and durability. Optional features such as a temperature monitoring indicator, leak-proof lid and ergonomic handles may be included to improve usability. The product should be affordable enough for adoption by small and marginal dairy farmers and suitable for village-level milk collection systems/centres.
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	Expected Solution A lightweight, insulated milk chilling can with a capacity of 30-40 litres that can maintain milk at safe temperatures for at least 6-12 hours without external power supply. The developed solution should reduce milk spoilage, improve milk quality during transportation, lower handling effort due to reduced weight and cost significantly less than conventional insulated stainless-steel chilling containers. The final prototype should be
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	durable, food-safe, easy to manufacture and suitable for widespread deployment in rural dairy supply chains.
Organization	Ministry of Fisheries, Animal Husbandry & Dairying
Department	Department of Animal Husbandry & Dairying
Category	Hardware
Theme	Agriculture , FoodTech & Rural Development
Dataset Link	Dairy temperature monitoring and milk quality datasets collected from local dairy cooperatives or

	experimental field trials.
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